



National University of Computer & Emerging Sciences, Karachi
Fall -2025 CS/AI/CY/SE-Department
Assignment- 3
24th October 2025



Course Code: CS2009	Course Name: Design & Analysis of Algorithms
Instructor Name:	
Date of Submission	4 th November 2025

Instructions:

Max Marks: 100

- Make your Assignments on Full Scale Papers, Typed Assignments will not be acceptable:
 - 20% penalty for 1 day late
 - 40% penalty for 2 days late
 - Submission not allowed afterwards
- 1) You are given a set of points in a 2D plane and you have to compute the convex hull of a given set of points using the Jarvis March (Gift Wrapping) algorithm, demonstrating how the orientation function helps identify the most counter clockwise point at each step. **[20 Marks]**

Given Points:

You are given the following six 2D points:

Label	Coordinates (x, y)
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A	(0, 3)
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B	(2, 2)
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C	(1, 1)
---	--------

D	(2, 1)
---	--------

E	(3, 0)
---	--------

F	(0, 0)
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Task:

1. Find the starting point — the point with the smallest x-coordinate (if tie, choose smallest y).
2. Use the orientation function to find, at each step, the most counter clockwise point from the current hull point.
3. Repeat until you return to the starting point.
4. List the convex hull points in counter clockwise order.

Orientation Function:

Use the orientation test to determine the turn direction for three points p,q,r:

$$\text{orientation}(p,q,r) = (q_x - p_x)(r_y - p_y) - (q_y - p_y)(r_x - p_x)$$

It tells us the direction of rotation from line segment $p \rightarrow q$ to $p \rightarrow r$:

Interpretation:

If orientation < 0 → Clockwise

If orientation > 0 → Counterclockwise

If orientation = 0 → Collinear

Step	Current Point (p)	Candidate (q)	Checking Point (r)	orientation(p, q, r)	More CCW?	New Candidate
1	F(0,0)	A(0,3)	B(2,2)	?	?	?
2	...	...	...	...	...	...
...						

Students must **fill in this table** for each iteration to determine which point becomes the next hull vertex.

Student Submission Requirements:

Students should:

1. Show orientation computations for each comparison.
 2. Fill the dry-run table completely.
 3. Draw a diagram labeling all six points and connecting the hull points in CCW order.
- 2) To understand and implement the computation of the KMP failure function (also called the prefix function) using both the naive (brute-force) method and the optimized (linear-time) KMP approach, and to compare their efficiency. **[20 Marks]**

The KMP failure function, $F(j)$, for a pattern $P[0..m-1]$ is defined as: "The size of the largest proper prefix of $P[0..j]$ that is also a proper suffix of $P[0..j]$." A proper prefix of a string is a prefix that is not equal to the entire string itself.

Tasks:

1. Using the above definition, write a brute-force algorithm to compute the failure function $F(j)$ for a given pattern P : $P = \text{"ababaca"}$
 - The algorithm should explicitly test all possible prefix and suffix matches for each position j .
 - Analyze the time complexity of this algorithm.
2. Write the optimized algorithm proposed by Knuth-Morris-Pratt (KMP) to compute the same failure function.
 - Clearly describe how it reuses previously computed values of $F(j)$.

- Analyze its time complexity.
3. For the pattern P compute the failure function F manually using both methods (brute force and optimized KMP).
 - Show all intermediate steps for each j .
 - Verify that both algorithms produce the same result.
 4. Compare the two algorithms in terms of:
 - Time complexity
 - Number of character comparisons
 - Reuse of computed information

Expected Output Format:

j	Substring $P[0..j]$	Longest Proper Prefix = Suffix	$F[j]$	3)
0	a	—	0	
1	ab	—	0	
2	aba	a	1	
...	...	...	...	

Understand, how these problems can be solved using dynamic programming and solve each of the following problem using an example. **[48 Marks]**

- a) The Coin Change Problem is a classic algorithmic problem that involves finding the number of ways to make change for a given amount of money (n) using a set of distinct coin denominations. Each coin denomination can be used unlimited times, and the goal is to determine how many combinations of coins can be used to make up the total amount. **Now solve for $S = \{1,5,6,8\}$, Desired Change is 13.**

[12 Marks]

- b) The Edit Distance, also known as the Levenshtein Distance, measures the similarity between two strings by calculating the minimum number of operations required to transform one series into another. The allowed operations are:

- i. Insertion: Add a character to the string.
- ii. Deletion: Remove a character from the line.
- iii. Substitution: Replace one character with another.

The goal is to find the minimum number of operations needed to equal two strings.

Now solve for $str1 = \text{"KITTEEN"}$, $str2 = \text{"SITTING"}$

[12 Marks]

- c) The Rod Cutting Problem is a classic optimization problem in computer science and mathematics. It involves finding the best way to cut a long rod into smaller pieces to maximize the total value of the details. Each piece of the rod has a length and an associated weight, and the goal is to determine the combination of cuts that yields the maximum value. **Now solve for $length[] = \{1, 2, 3, 4, 5, 6, 7, 8\}$, $price[] = \{1,5, 8, 9, 10, 16, 18, 20\}$, Rod Length: 8.**

[12 Marks]

d) Given an input string s and a dictionary of words wordDict , determine if s can be segmented into a space-separated sequence of one or more words from wordDict . Words in wordDict can be reused. **Now solve for $\text{wordDict} = \{\text{i, like, ice, cream, icecream, mobile, apple}\}$, Input: ilikeapple .**

[12 Marks]

- 4) Take any real-world daily life task and write an algorithm to solve that using dynamic programming approach that is mapped on one of the problems above. **[12 Marks]**

For example: A university student wants to register for courses this semester.

Each course:

- gives a number of credits (academic value or “benefit”), and
- requires a certain number of study hours per week (effort or “cost”).

However, the student only has a limited number of study hours they can dedicate weekly (say, due to part-time work, health, or personal limits).

So, the student must select the best combination of courses that:

- maximizes total earned credits (academic value),
- while keeping the total study hours within the weekly limit.